

DELETE Basics

DELETE removes rows from a table.

General syntax:

```
DELETE FROM table_name;
```

Unlike DROP, the table structure remains.

- Finds matching rows.
- Locks those rows.
- Marks them for deletion.
- Writes changes to the transaction log.
- Frees the space later (depends on the storage engine).

DELETE Without WHERE

```
DELETE FROM Students;
```

Every row is deleted.

DELETE vs TRUNCATE

Feature	DELETE	TRUNCATE
Removes	Selected rows or all rows	All rows
WHERE	✓ Yes	✗ No
Logged	Row by row	Minimal logging
Rollback (InnoDB transaction)	✓ Yes	Depends on DBMS; MySQL treats it as DDL
AUTO_INCREMENT reset	✗ No	✓ Yes (MySQL)
Speed	Slower	Faster

DELETE Using Subqueries

Example:

```
DELETE FROM Employees
WHERE EmployeeID IN (
    SELECT EmployeeID
    FROM FormerEmployees
);
```

Cascading Delete

Suppose

- **Parent Table:** Departments
- **Child Table:** Employees

Foreign key:

```
FOREIGN KEY (DepartmentID)
REFERENCES Departments(DepartmentID)
ON DELETE CASCADE
```

Now

```
DELETE FROM Departments
WHERE DepartmentID = 10;
```

Automatically deletes

- Department 10
- Every employee belonging to Department 10

Without ON DELETE CASCADE

The deletion fails if child rows exist.

Large applications often don't actually delete rows.

Instead:

```
UPDATE Students
SET is_deleted = TRUE
WHERE StudentID = 1;
```

Then all queries become:

```
SELECT *
FROM Students
WHERE is_deleted = FALSE;
```

Benefits:

- Recover deleted data
- Audit history
- Compliance

- No accidental permanent deletion

Recovering Deleted Data

Transactions

```
START TRANSACTION;  
DELETE FROM Students  
WHERE StudentID = 1;
```

Oops!

Recover:

```
ROLLBACK;
```

Everything is restored.

Save permanently:

```
COMMIT;
```

Backups

If the transaction is already committed, recovery usually requires:

- Database backups
- Binary logs (MySQL)

Can DELETE be rolled back?

Yes, if executed inside a transaction (for transactional storage engines like InnoDB) and before COMMIT.

What is ON DELETE CASCADE?

Automatically deletes child rows when the parent row is deleted.

What is Soft Delete?

Instead of removing rows, mark them as deleted using a flag such as `is_deleted`.